FORMAL PROFESSIONAL SCOPE STATEMENT

Licensed Professional Activities

Chemical Engineering and Chemical Consulting

Diwan of Excellence Engineering Consultations

License No.	5100001788
Professional discipline	Chemical Engineering
Activity codes	711032 and 749073
Purpose	Statement of services the office may provide within the attached license

This statement describes the services the office may provide under the attached license only. Code 711032 is applied solely to the Chemical Engineering component, while code 749073 covers Chemical Consulting Activities. This statement does not add any activity or discipline that is not stated on the license.

License Basis

Code	Activity Wording on the License	Application to the Office
711032	Engineering activities for electrical projects, excavation, security and safety, and chemical projects (English rendering of the Arabic license wording)	Engineering services strictly within Chemical Engineering: studies, design, calculations, drawings, specifications, review, supervision and commissioning associated with chemical processes.
749073	Chemical consulting activities	Professional advice, studies, assessment and technical interpretation concerning chemistry, materials, products, analysis, testing and chemical processes.

The discipline restriction is essential. Although the umbrella wording of code 711032 lists several fields, the professional discipline recorded for the office is Chemical Engineering. Accordingly, this statement does not claim standalone electrical design, excavation/geotechnical work, security systems, or independent fire-protection engineering. Safety services are limited to chemical and process safety within Chemical Engineering.

How the Two Codes Work Together

Where the service is an engineering design, calculation, drawing or supervision activity, its primary basis is code 711032. Where it is a chemical study, material assessment, analytical plan or interpretation of results, its primary basis is code 749073. Many assignments use both codes—for example, diagnosing a process problem through analytical data and then modifying the process design or operating conditions.

1. Chemical Engineering and Process Engineering Consulting

- Conceptual and technical studies, technical feasibility studies, and selection of process routes and technologies.
- Preparation of basis of design, design criteria, process philosophies and operating requirements.
- Design, development, optimization and technical comparison of chemical and industrial processes.
- Mass and energy balances; thermodynamic, heat-transfer, fluid-flow and pressure calculations.
- Process simulation, performance optimization, debottlenecking and definition of suitable operating conditions.
- Preparation and review of process flow, utility flow, and piping and instrumentation diagrams (PFD, UFD and P&ID).
- Process design for reaction, mixing, separation, distillation, absorption, extraction, filtration, drying, crystallization and thermal treatment.
- Design of laboratory, pilot and semi-industrial systems, including scale-up from laboratory to industrial operation.
- Process sizing and preparation of data sheets and technical specifications for reactors, vessels, columns, exchangers, pumps and other process equipment.
- Definition of utility requirements and consumption, including water, steam, cooling, heating, air, vacuum and industrial gases.
- Definition of process measurement, monitoring, control, alarm and interlock requirements from a Chemical Engineering perspective.
- Improvement of energy and material efficiency and reduction of losses, emissions and waste from a process perspective.

2. Process Safety and Chemical Hazards

This section covers chemical process safety as part of Chemical Engineering; it does not constitute standalone security, fire-protection or electrical-safety engineering.

- Chemical hazard identification and design/operability reviews using HAZID, HAZOP, What-if and checklist methods as appropriate.
- Assessment of uncontrolled reactions, thermal decomposition, material incompatibility and runaway reaction hazards.
- Review of safe operating limits, protection layers, alarms and shutdown requirements from a process perspective.
- Preparation or review of pressure-relief loads and process requirements for vent and flare systems.
- Screening-level consequence and dispersion assessments, risk ranking and mitigation recommendations.

- Requirements for safe storage, segregation, handling and internal transfer of chemicals.
- Incident, deviation and root-cause analysis, with corrective and preventive recommendations.
- Preparation of project-specific process-safety and management-of-change procedures and elements.

3. Chemical, Materials and Product Consulting

- Development and improvement of chemical products and formulations, including raw-material and additive selection and technical ratios.
- Preparation of raw-material and product specifications, acceptance limits and technical quality-control plans.
- Assessment of alternative materials, supplier comparison, and technical review of material and product certificates.
- Chemical compatibility studies and selection of construction materials for temperature, pressure and composition conditions.
- Assessment of corrosion and chemical degradation and selection of materials, linings, coatings and protection methods.
- Consulting on polymers, plastics, rubber, resins, composites, fibers and fillers.
- Optimization of extrusion, injection molding, mixing, forming, processing and curing, and correlation with final properties.
- Consulting on catalysts, adsorbents, porous materials, catalytic processes and catalyst deactivation.
- Technical assessment of paints, coatings, inks, adhesives and specialty chemicals.
- Stability, storage and technical shelf-life studies, including preparation of test plans and review of results.
- Investigation of material or product failure and changes in color, odor, viscosity, strength or other properties.
- Identification of contamination, impurities and degradation sources associated with raw materials, storage, manufacturing or use.
- Preparation and technical review of chemical-safety information, safety data sheets and handling requirements.

4. Analytical Consulting, Test Planning and Results Interpretation

Yes—the scope is broad and is not limited to GC or HPLC. The office may prepare analytical and testing plans for gases, liquids, solids, powders, polymers, composites, petroleum and petrochemical products, catalysts, coatings, water, process wastes and other relevant materials. The selected test is driven by the technical question, sample type and required decision—not by a specific instrument.

- Translate an industrial problem or quality question into analytical objectives and clear decision criteria.
- Design analytical and characterization programs, including primary and confirmatory tests and their sequence.
- Prepare sampling plans covering number, representativeness, preservation, transport, preparation and chain of custody where required.
- Select the applicable test method or standard and the suitable instrument or technique, with technical justification.
- Define acceptance limits, range, sensitivity, detection and quantification limits, accuracy, precision and required uncertainty.
- Prepare or review analytical methods and assess fitness for purpose, including method verification or validation plans.
- Prepare laboratory scopes of work, technically compare laboratory quotations, and define report and raw-data requirements.
- Review quality-control data, blanks, reference materials, replicates, calibration and overall result validity.
- Identify implausible or conflicting results and specify suitable confirmatory work.
- Integrate results from multiple techniques and correlate them with composition, structure, properties, manufacturing and operation.
- Compare findings against contractual specifications, standards and product requirements, and identify deviations.
- Issue a technical or engineering advisory report containing interpretation, conclusions, recommendations and action plans.

Execution boundary: the office may design the program, select methods and laboratories, review data quality, interpret results and issue an advisory report. Physical measurements and any test report or certificate of analysis represented as an accredited laboratory document must be issued by a competent/accredited laboratory where accreditation is required by regulation or the client.

Categories of Analytical and Testing Techniques — Non-exhaustive Examples

Category	Example Techniques and Tests
Separation, chromatography and mass spectrometry	GC, GC-MS, HPLC/UPLC, LC-MS, IC and GPC/SEC for composition, impurities and molecular-distribution questions.
Molecular and spectroscopic analysis	FTIR, Raman, UV-Vis, NIR and NMR for identification, functional groups, composition and chemical change.
Elemental and chemical analysis	ICP-OES, ICP-MS, AAS, XRF, CHNS/O, titration, wet chemistry, ions and moisture.
Thermal analysis and stability	DSC, TGA, DMA, TMA, calorimetry, stability, decomposition and thermal-reaction testing.
Structure, phases, morphology and surfaces	XRD, SEM-EDS, optical microscopy, BET, porosity, surface area and surface-analysis techniques.

Category	Example Techniques and Tests
Physical, mechanical and rheological properties	Density, viscosity, flow, particle size, tensile, compression, impact, hardness, MFI and rheology.
Gas, petroleum and petrochemical testing	Gas composition, calorific value, sulfur, distillation curves, vapor pressure, flash point, viscosity, density, water and sediment.
Corrosion, coatings and electrochemistry	Immersion, polarization, EIS, salt spray, coating thickness, adhesion and chemical resistance to the applicable method.
Water, process streams and wastes	pH, conductivity, TOC, COD, BOD, ions, metals, solids and treatment-performance indicators.

5. Materials and Markets the Office May Serve

- Oil and gas, refining, petrochemicals, feedstocks, intermediates and finished products.
- Industrial gases, hydrogen, natural gas, carbon dioxide, ammonia, process gases and emissions.
- Polymers, plastics, rubber, resins, composites, fibers and fillers.
- Catalysts, adsorbents, porous materials, and advanced and functional materials.
- Paints, coatings, inks, adhesives and specialty chemicals.
- Water, industrial wastewater, sludge, wastes, treatment chemicals and treatment processes.
- Metals, ceramics, salts and construction materials from a chemical and materials perspective.
- Energy materials, batteries, electrodes and chemical energy-storage and conversion materials.
- Manufacturing plants, R&D centers and laboratories for scale-up, troubleshooting, quality improvement and process improvement projects.

6. Standards, Technical References and Relevant Bodies

The office may use the applicable standard as a reference for design, testing, acceptance or review. Its role may include confirming the applicable edition, developing a requirements matrix, preparing the test plan, reviewing evidence and results, and identifying technical gaps. Use of a standard does not make the office the certificate issuer.

Reference	Typical Area	Services the Office May Provide
ISO / ISO/IEC	International technical, management-system, laboratory-competence and conformity-assessment standards.	Identify project-related technical requirements, conduct technical gap reviews, prepare technical evidence and support readiness, including advisory support relating to ISO/IEC 17025.
ASTM International	Test methods and specifications for chemicals, petroleum, polymers, metals and industrial materials.	Select the method and edition, prepare sampling and test plans, define acceptance criteria, review application and results, and interpret findings.
SASO	Saudi standards, technical regulations and conformity-assessment procedures in the Kingdom.	Identify the applicable regulation or standard, prepare compliance matrices and technical files, and review reports before submission to the competent body.

Reference	Typical Area	Services the Office May Provide
GSO	GCC standards and technical regulations for Gulf Cooperation Council member states.	Identify Gulf requirements, compare them with product or material characteristics, and prepare technical assessments and gap reviews.
API / ASME	Oil and gas, process, piping, equipment and pressure-equipment references.	Prepare process/material requirements and specifications and review manufacturer/vendor documents within Chemical Engineering scope.
AMPP / NACE	Corrosion, materials protection and coatings.	Develop corrosion strategies, support material/coating selection, prepare test plans and review results within the chemical scope.
EN / BS / DIN	European, British and German standards for materials, equipment and test methods.	Apply the contractually required reference, prepare technical comparisons, and review documentary and test-result conformity.
Owner or client specifications	Project-specific and proprietary requirements when supplied and authorized for use.	Translate them into technical requirements, test plans, and bid/vendor reviews without implying owner approval or prequalification of the office.

7. Conformity, Certification and Accreditation — The Regulatory Distinction

The office is an engineering/chemical consultancy. It may prepare evidence, perform technical assessments and support readiness; however, status as a testing, inspection, certification or accreditation body does not arise from this license alone.

Service or Document	Status	Clarification
Identify the applicable standard or technical regulation	**Within scope**	Prepare a register of requirements, editions and references applicable to the material, product or project.
Test plan based on ISO, ASTM, SASO or GSO	**Within scope**	Define samples, methods, tests, acceptance limits and reporting requirements.
Technical compliance matrix or gap analysis	**Within scope**	Compare design, documents or results with standard or regulatory clauses and identify gaps.
Review laboratory reports, CoA, MTC and SDS	**Within scope**	Technically review content, results and conformity with specified requirements.
Prepare technical file and support readiness	**Within scope**	Compile and organize technical evidence and support closure of findings before submission to the competent body.
Advisory report interpreting results and recommending action	**Within scope**	The office may issue a technical/engineering advisory report within its licensed discipline.
ISO management-system or product certificate	**Not issued by the office in this capacity**	ISO does not issue certificates; certification is performed by an independent certification body under the applicable scheme.
Saudi or GCC product conformity certificate or mark	**Not issued by the office in this capacity**	Issued through the competent authority or an authorized conformity-assessment body under the applicable program or regulation.
Test report or certificate of analysis represented as accredited	**Not issued by the office in this capacity**	Issued by a competent/accredited laboratory for the required scope; the office may review and interpret it.
Accreditation of a laboratory, inspection body or certification body	**Not within the office's authority**	Accreditation is granted by the competent accreditation body; in Saudi Arabia, SAAC accredits conformity-assessment bodies.

Service or Document	Status	Clarification
Statutory inspection report or permit requiring special authorization	**Separate authorization required**	The office may provide technical support, but the statutory document must come from the authorized body or person.

Who Does What?

Party	Typical Role
Consulting office	Identifies requirements; prepares studies and plans; reviews and interprets evidence and results; provides recommendations and readiness support.
Laboratory	Performs measurements and tests and issues test reports or certificates of analysis within its competence/accreditation scope.
Inspection body	Performs inspection and issues inspection outputs within its accreditation or authorization scope.
Certification body	Performs conformity assessment and issues a certificate when scheme requirements are fulfilled.
Saudi Accreditation Center (SAAC)	Accredits laboratories, inspection bodies, certification bodies and other conformity-assessment bodies in Saudi Arabia.
SASO / GSO and regulators	Develop or adopt relevant standards, technical regulations and programs and establish conformity procedures according to their mandates.

8. Project, Vendor and Commissioning Support

- Prepare scopes of work, technical specifications, data sheets, and tender and contract technical requirements.
- Prepare technical schedules and quantities related to Chemical Engineering equipment and materials.
- Technically evaluate bids and compare technologies, suppliers and equipment, including deviations and exceptions.
- Review manufacturer and vendor designs and documents against the basis of design and process requirements.
- Review project inspection and test plans from a technical perspective and witness tests when contracted, without claiming accredited-inspection status.
- Provide technical supervision of manufacturing, installation and implementation within Chemical Engineering scope.
- Supervise pre-commissioning, commissioning, performance testing and technical process acceptance.
- Prepare operating, shutdown, emergency, inspection, technical follow-up and troubleshooting procedures.
- Provide project- or process-specific technical training and knowledge transfer as part of the consulting service.

9. Coverage Map for the Two Activity Codes

Service Group	Primary Code Basis	How It Is Covered
Studies, design, calculations and drawings	711032	Engineering work within the Chemical Engineering discipline.
Process, equipment, utilities and scale-up engineering	711032	Engineering design, review, optimization and supervision.
Process safety and chemical hazards	711032 + 749073	Engineering and chemical study tied to the process—not standalone safety work outside the discipline.
Materials, products, formulations, polymers and catalysts	749073; 711032 when an engineering application is included	Chemical consulting, with engineering coverage where connected to process or design.
Analytical/test plans and results interpretation	749073; 711032 where results affect design or operation	Broad analytical consulting for all material states, without restriction to a particular instrument.
Standards, technical conformity and readiness support	749073 + 711032 according to subject	Requirements identification and evidence review—not certification or accreditation.
Vendor evaluation, supervision and commissioning	711032	Engineering project services within Chemical Engineering.
Troubleshooting, failure investigation and recommendations	711032 + 749073	Combines chemical analysis and interpretation with an engineering solution.

10. Submission-ready Formal Statement

The office provides Chemical Engineering and Chemical Consulting services, including studies; process design, development and optimization; process safety and chemical-hazard studies; design of reactors and laboratory, pilot and semi-industrial systems; preparation of calculations, drawings and technical specifications; consulting on materials, products, chemical formulations, polymers, composites, catalysts, coatings and corrosion; preparation of sampling, analytical and testing plans for all material states; selection of applicable methods and standards; review of data quality and interpretation of results; technical conformity assessments, gap analyses, technical files and readiness support for competent bodies; investigation of operational problems and failures; technical evaluation of bids and suppliers; and technical supervision, commissioning and performance testing, all strictly within the Chemical Engineering discipline and activity codes 711032 and 749073 stated on the license, without claiming status as an accredited laboratory, inspection body, certification body or accreditation body.

Basis and Limits of This Statement

This statement is an expanded professional interpretation of the attached license. It does not amend the license wording or constitute advance regulatory approval for a particular project. Any sector-specific permit, authorization or accreditation remains applicable according to the product, project and contract.

- Attached professional engineering company license No. 5100001788, Chemical Engineering discipline, activity codes 711032 and 749073.
- Saudi Standards, Metrology and Quality Organization (SASO) — Responsibilities
- Saudi Accreditation Center (SAAC) — Accreditation services and FAQ
- ISO — Certification and conformity assessment
- GCC Standardization Organization (GSO) — GCC standards